

NEURAL NETWORKS

Course:
Artificial Intelligence – Understanding opportunities and
limitations across disciplines
10th January 2025,
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Mohamed Elsafty: Topic 1, 2, 4, 7, 8
Vignesh Mallya : Topic 3, 4, 5, 6

WHAT ARE NEURAL NETWORKS?

NEURAL NETWORKS

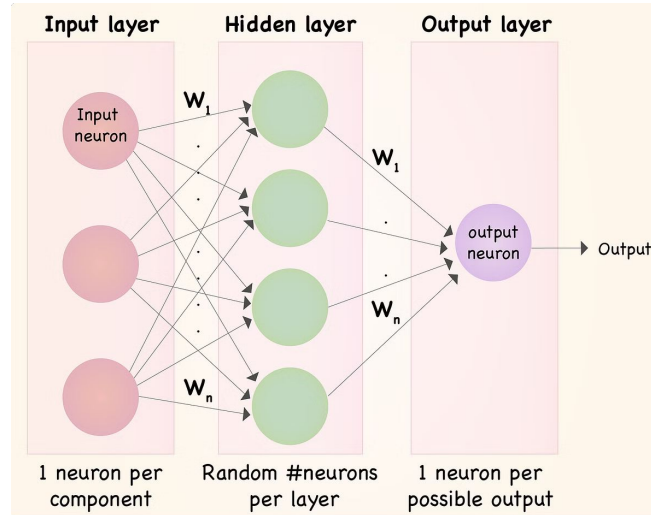
WHAT ARE NEURAL NETWORKS?

Algorithms that Learn

Neural networks are algorithms designed to identify patterns within data.

Interconnected Neurons

These algorithms are composed of interconnected 'neurons', organized into layers.



Connection Adjustment

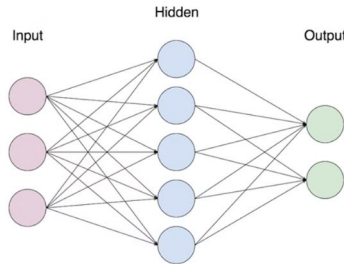
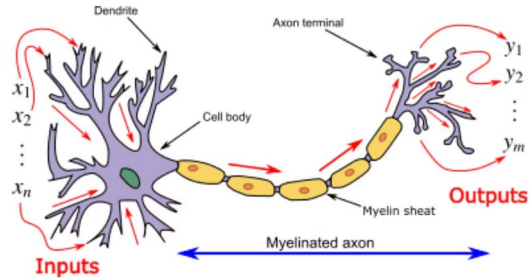
The connections between neurons are dynamically adjusted based on the input data to optimize predictive capabilities.

Function Approximators

Essentially, they are sophisticated function approximators that learn through iterative examples.

NEURAL NETWORKS

BIOLOGICAL INSPIRATION



1

Biological Neural Networks

Brains contain billions of interconnected neurons; the strength of these connections changes through learning.

2

Mimicking the Process

Artificial networks mimic this by adjusting weights during training.

3

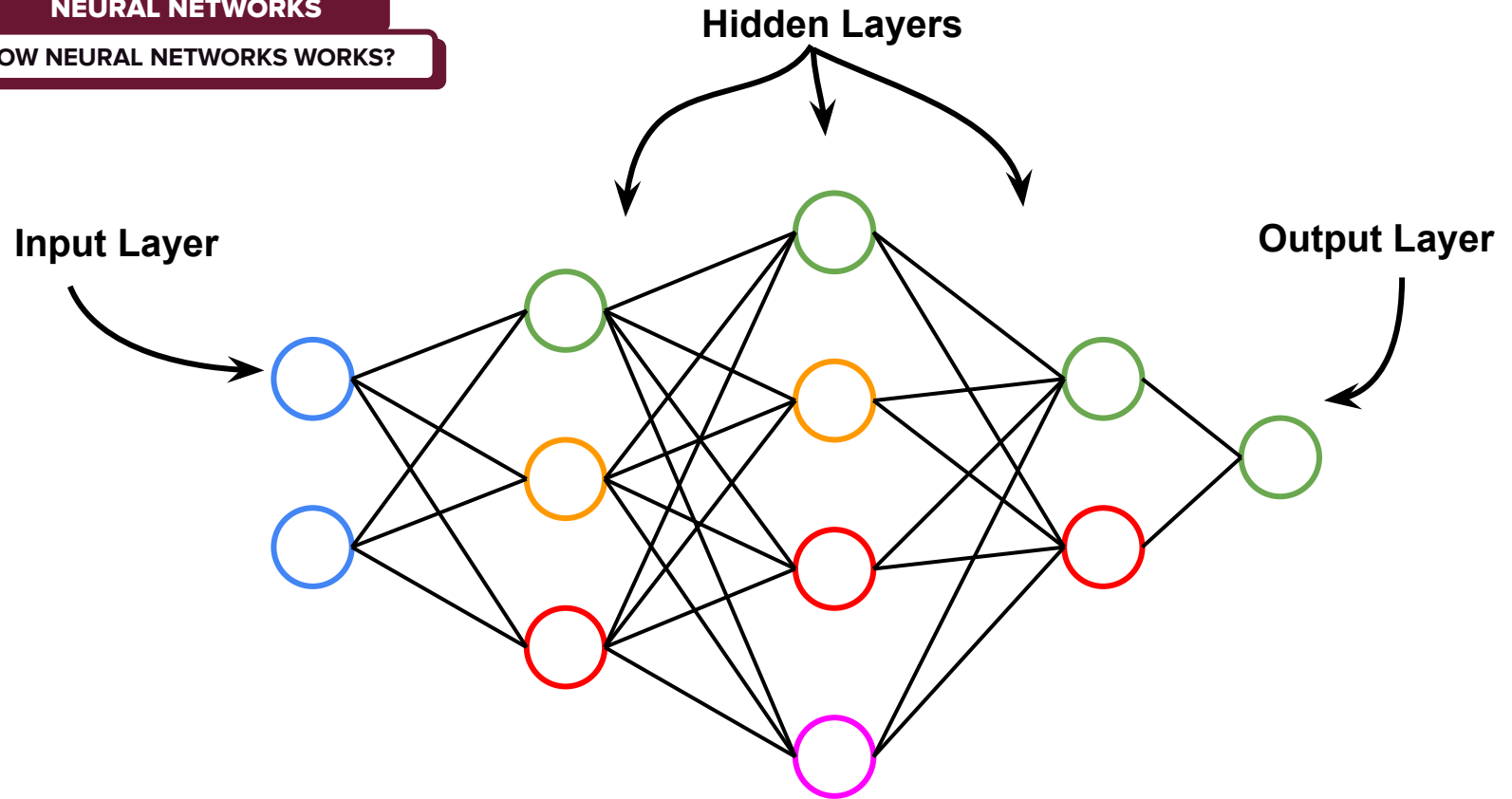
Flexibility and Power

This brain-inspired architecture gives neural networks their flexibility and power.

HOW NEURAL NETWORKS WORK?

NEURAL NETWORKS

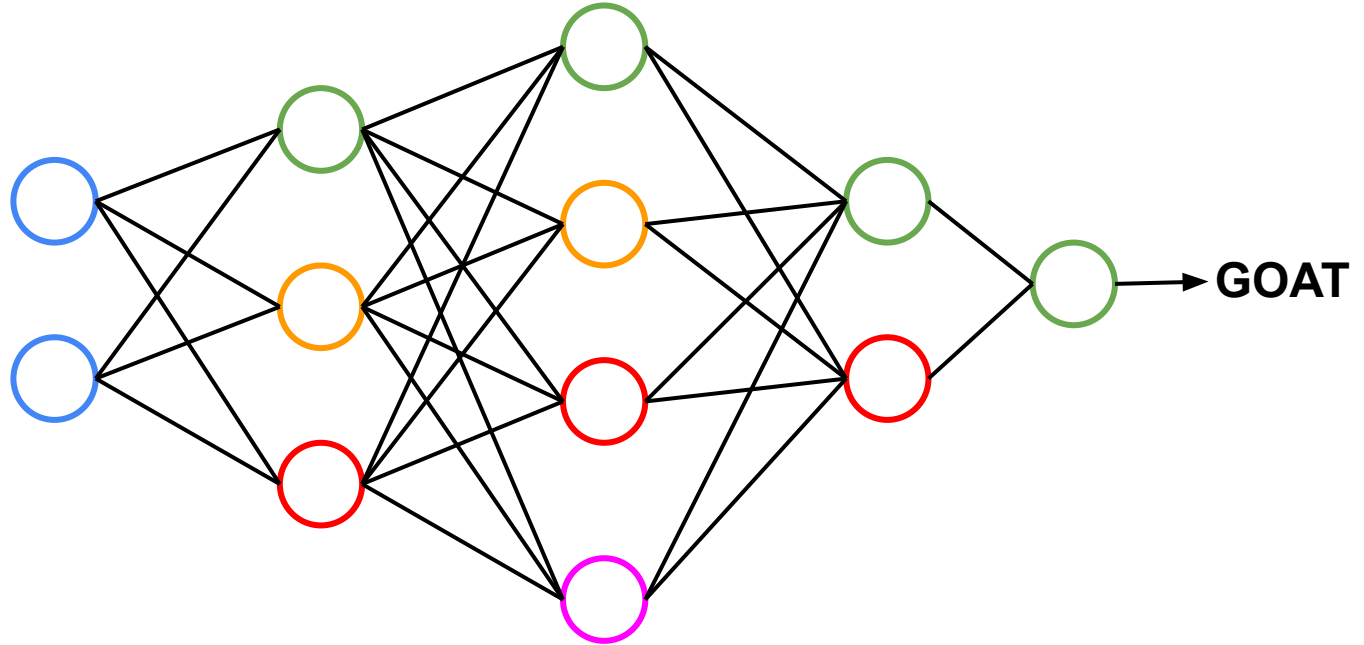
HOW NEURAL NETWORKS WORKS?



NEURAL NETWORKS

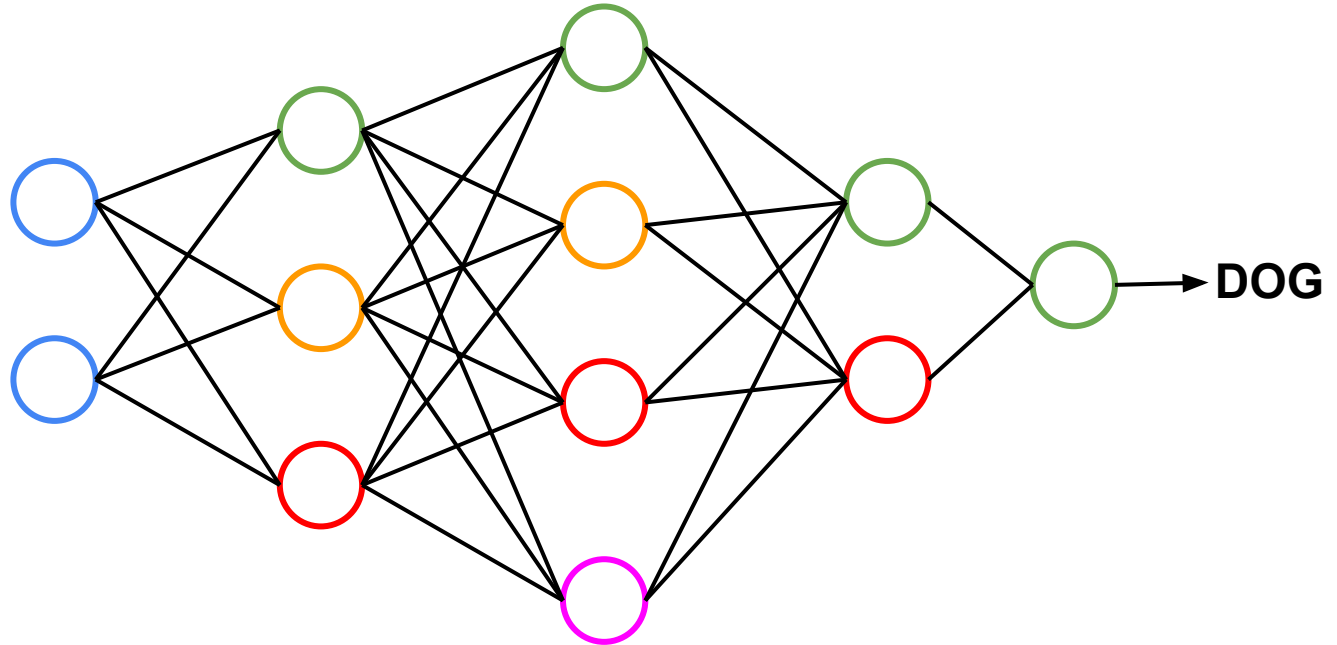
HOW NEURAL NETWORKS WORKS?

Forward Propagation



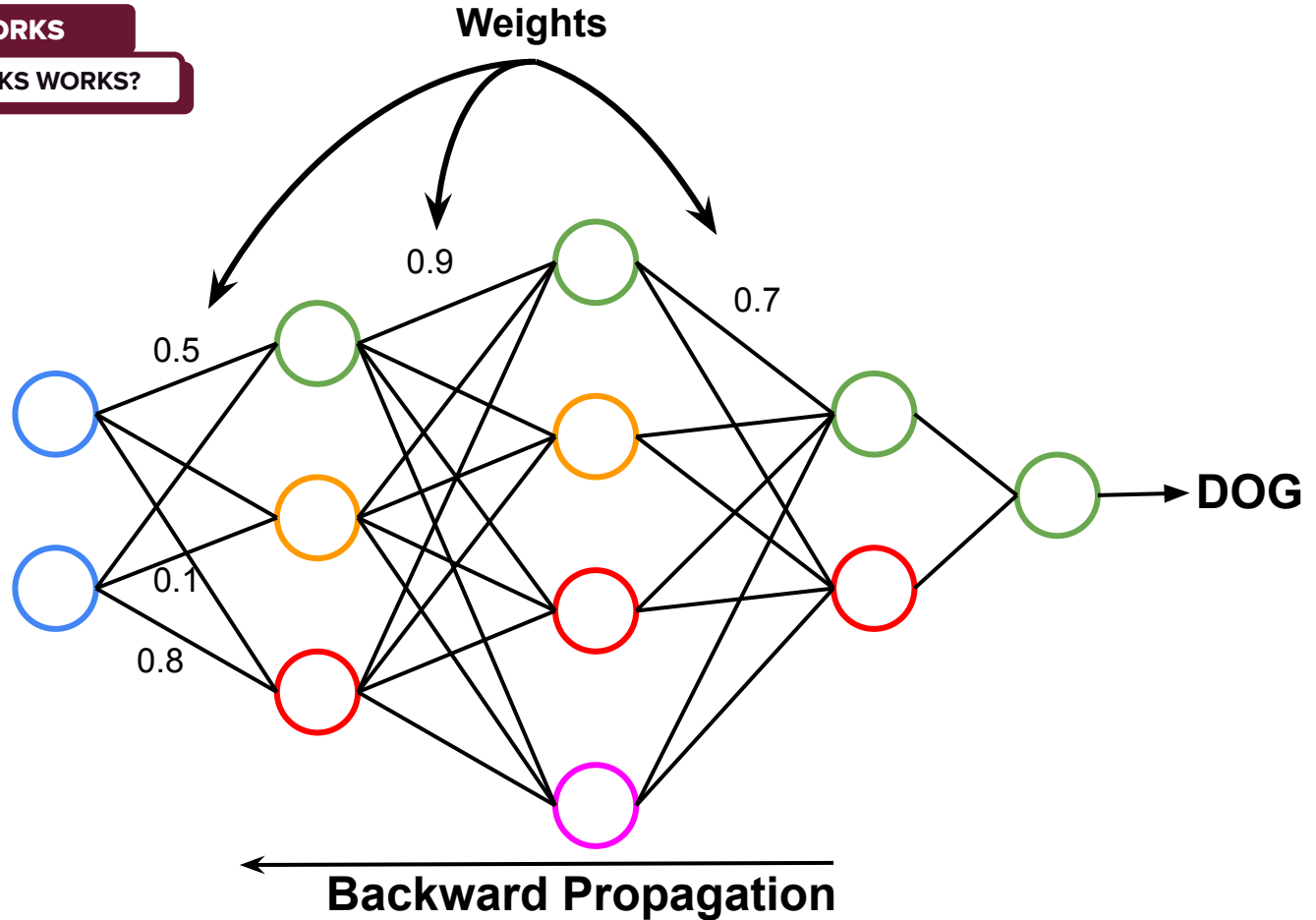
NEURAL NETWORKS

HOW NEURAL NETWORKS WORKS?



NEURAL NETWORKS

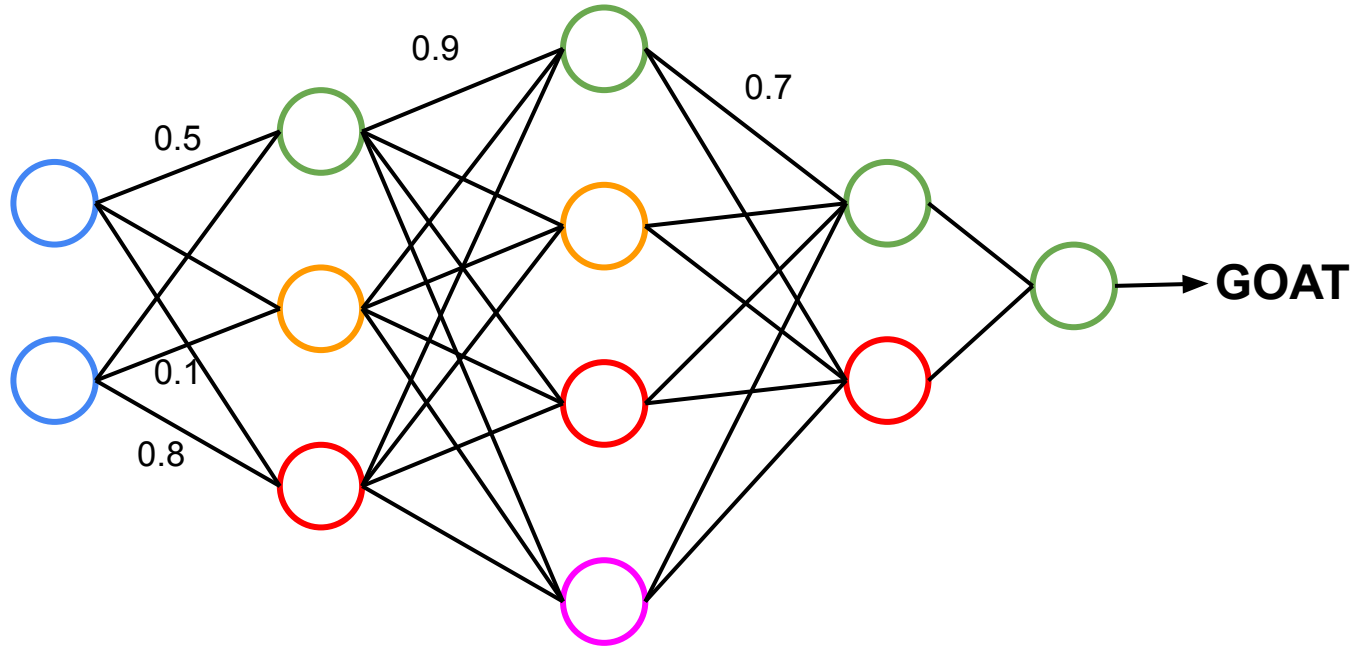
HOW NEURAL NETWORKS WORKS?



NEURAL NETWORKS

HOW NEURAL NETWORKS WORKS?

$$\text{Output} = \text{Activation Function}(\sum(\text{Input} \times \text{Weight}) + \text{Bias})$$



HOW TO TRAIN NEURAL NETWORKS?

Supervised Learning

Training with labeled data to map inputs to desired outputs.

Unsupervised Learning

Training without labels, focusing on identifying patterns and clusters in data.

Reinforcement Learning

Learning through interaction with an environment using rewards and penalties to optimize strategies

TYPES OF NEURAL NETWORKS

NEURAL NETWORKS

FEEDFORWARD NEURAL NETWORKS

FEEDFORWARD NEURAL NETWORKS (FNN)

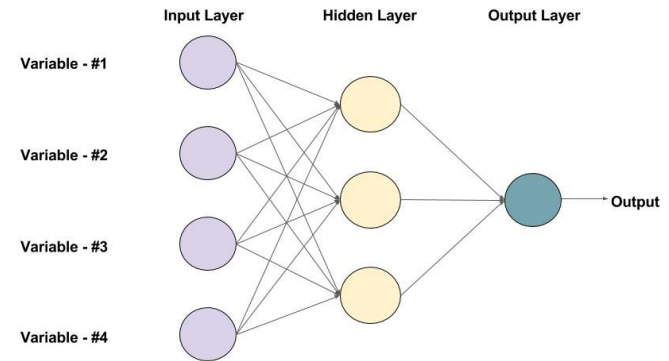
The simplest type of NN where data flows in one direction: from input to output. Consists of an input layer, hidden layers, and an output layer. No loops or cycles.

❖ Use Cases of FNNs

- **Classification:**
Predicting class labels based on input features, commonly used for simple tasks where relationships are not too complex.
- **Regression:**
Predicting continuous output values, useful in tasks where input features are related to a continuous target variable.

❖ Advantages

Easy to implement. Suitable for simple pattern recognition tasks.



An example of a Feed-forward Neural Network with one hidden layer (with 3 neurons)

<https://learnopency.com/understanding-feedforward-neural-networks/>

NEURAL NETWORKS

CONVOLUTIONAL NEURAL NETWORKS

CONVOLUTIONAL NEURAL NETWORKS (CNN)

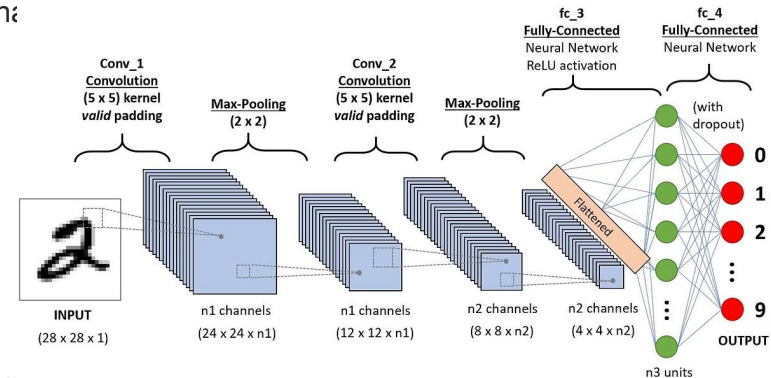
Specialized NNs designed for processing grid-like data, such as images. Employ convolutional layers that detect spatial hierarchies and patterns like edges, shapes, and textures. Includes pooling layers for dimension reduction and feature extraction.

❖ Use Cases of CNNs

- **Image Recognition:**
Identifying objects, patterns, or features within images.
- **Object Detection:**
Detecting and locating objects within images.
- **Video Analysis:**
Analyzing video data by processing frames sequentially or in parallel to capture temporal relationships.

❖ Advantages

High accuracy in image-related tasks. Efficient handling of large-scale data.



<https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>

RECURRENT NEURAL NETWORKS

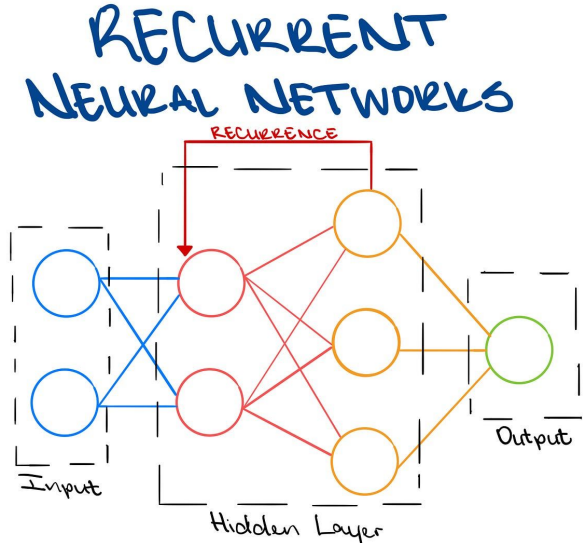
RNNs are particularly well-suited for processing sequential data, such as text or audio. They have internal memory to remember past inputs, allowing them to understand context and dependencies in data.

❖ Use Cases of RNNs

- **Natural Language Processing (NLP):**
Language modeling, Text generation, Machine translation
- **Speech Recognition:**
Processing audio data for recognizing speech patterns
- **Time Series Prediction:**
Modeling temporal dependencies for forecasting

❖ Advantages

RNNs excel at capturing long-term dependencies, making them powerful tools for understanding complex sequential data.



<https://towardsdatascience.com/introducing-recurrent-neural-networks-f359653d7020>

GENERATIVE ADVERSARIAL NETWORKS

GANs are a type of generative model that learns to create new data samples that resemble the training data.

Generator: Creates fake data.

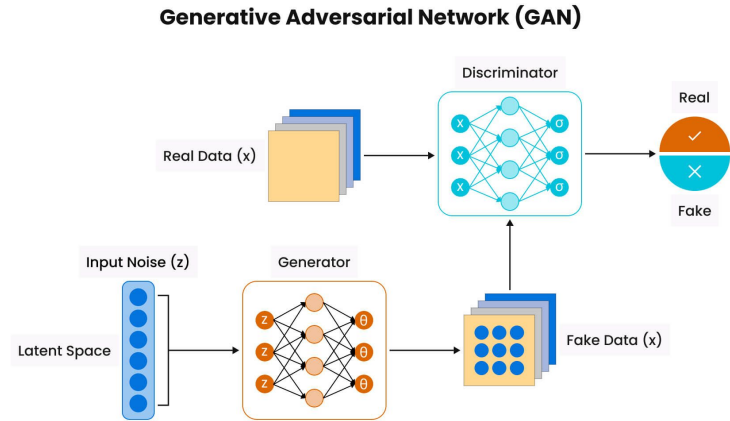
Discriminator: Tries to distinguish between real and fake data.

❖ Use Cases of GANs

- **Image Generation:**
Creating realistic, high-resolution photos.
- **Data Augmentation:**
Generating synthetic data to augment training data.
- **Creative Applications:**
Generating artwork or music.

❖ Advantages

GANs are known for their ability to generate high-quality, realistic data samples, surpassing traditional generative methods.

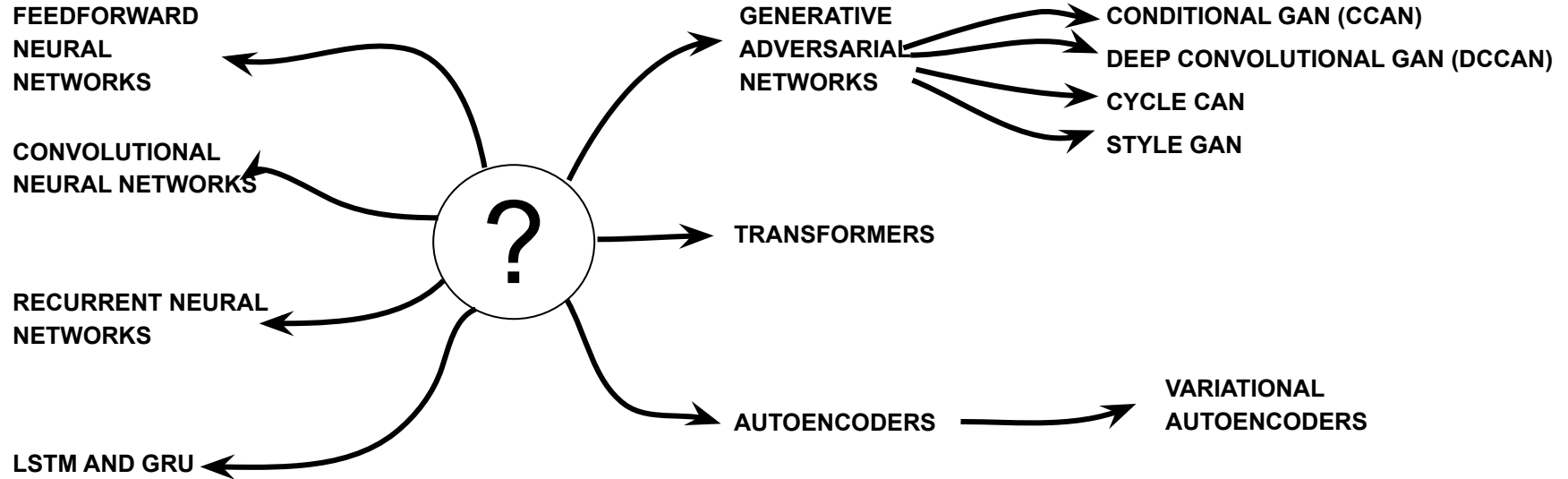


<https://www.solulab.com/generative-adversarial-network/>

NEURAL NETWORKS

TYPES OF NEURAL NETWORK

DIFFUSION MODELS



ADVANTAGES & APPLICATIONS

- ❖ **Adaptability:** Learns complex or undefined relationships between inputs and outputs.
- ❖ **Pattern Recognition:** Excels in tasks like audio/image recognition and natural language processing.
- ❖ **Parallel Processing:** Processes multiple tasks simultaneously for faster computations.
- ❖ **Non-Linearity:** Models complex relationships using non-linear activation functions.

NEURAL NETWORKS

APPLICATIONS

COMPUTER VISION

Neural networks analyze images and videos to detect objects, recognize faces, and enable advanced visual tasks.

NATURAL LANGUAGE PROCESSING

Neural networks process and understand human language for translation, search, and conversational AI.

SPEECH RECOGNITION

They convert spoken language into text, enabling voice assistants and real-time transcription.

RECOMMENDATION ENGINES

They predict user preferences to provide personalized suggestions for content or products.



ETHICAL IMPLICATION, LIMITATIONS & CHALLENGES

ETHICAL IMPLICATIONS AND LIMITATIONS



Autonomy and Individuality

Neural networks are increasingly used in AI systems that make decisions impacting hiring, lending, and healthcare. Over-reliance on these automated systems risks diminishing human autonomy in decision-making.



Responsibility Gaps

Determining accountability for decisions made by neural networks remains challenging. This is particularly apparent in cases like autonomous vehicle accidents where responsibility is unclear.



Societal Implications

Widespread AI adoption has the potential to significantly disrupt various aspects of society, including job markets, privacy, and socio-economic structures. Automation, for instance, may lead to widespread unemployment.

BLACK-BOX NATURE AND CHALLENGES

Black-Box Nature

NNs are often criticized for their opacity—difficult to interpret or explain how they arrive at decisions. This "black-box" nature undermines trust, especially in high-stakes domains like healthcare and law enforcement.

Computational Costs

Training deep NNs requires vast computational resources, leading to high energy consumption and carbon footprint. Example: Training GPT-3 required 1,287 MWh of electricity and emitted 552 metric tons of CO₂.

Bias in Training Data

If training data is biased, NNs perpetuate and even amplify those biases. Example: Facial recognition systems often perform poorly on minority groups due to underrepresentation in training datasets.



THANK YOU!

1. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *_Deep Learning_*. MIT Press
2. LeCun, Y., Bengio, Y., & Hinton, G. (2015). "Deep learning." *_Nature_*, 521(7553), 436-444.
3. Domingos, P. (2015). *_The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World_*. Basic Books.
4. Strubell, E., Ganesh, A., & McCallum, A. (2019). "Energy and policy considerations for deep learning in NLP." *_arXiv preprint arXiv:1906.02243_*.
5. O'Neil, C. (2016). *_Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy_*. Crown Publishing Group.
6. Aston Zhang, Zachary C. Lipton, Mu Li, & Alexander J. Smola. (2023). *Dive into deep learning*. D2L.ai. <https://d2l.ai/index.html>
7. Amazon Web Services. (n.d.). *What is a neural network?* <https://aws.amazon.com/what-is/neural-network/>
8. GeeksforGeeks. (2024). *Neural networks: A beginner's guide*. <https://www.geeksforgeeks.org/neural-networks-a-beginners-guide/>
9. Coursera. (2024, April 10). *Neural network examples, applications, and use cases*. <https://www.coursera.org/articles/neural-network-example>